One-step forward construction

(see Holtzman, Beauvallet, Krauth: https://arxiv.org/abs/2606.12161)

Here we look at Eqs. (13) and (14), but exchange the integration in x_β and in $x_{\tau+\Delta\tau}$. We first integrate out over x_β all the terms that depend on it.

```
Integrate[Exp[-xbeta^2 / (2 sigmaR^2)] Exp[-(xtau - xbeta)^2 / (2 (beta - tau))]
  Exp[-(xtauplus - xbeta)^2 / (2 (beta - tau - Deltau))] Exp[(xbeta - xzero)^2 / (2 beta)]
  Exp[(xtau - xbeta)^2 / (2 (beta - tau))], {xbeta, -Infinity, Infinity}]
```

Out[•]=

$$\frac{e^{\frac{\text{sigmaR}^2\,(\text{xtauplus}-\text{xzero})^2-(\text{Deltau}+\text{tau})\,\text{xzero}^2+\text{beta}\,\left(-\text{xtauplus}^2+\text{xzero}^2\right)}{2\left(\text{beta}^2-\text{beta}\,(\text{Deltau}+\text{tau})+\text{sigmaR}^2\,(\text{Deltau}+\text{tau})\right)}}\sqrt{2\pi}}{\sqrt{-\frac{1}{\text{beta}}+\frac{1}{\text{sigmaR}^2}+\frac{1}{\text{beta}-\text{Deltau}-\text{tau}}}} \quad \text{if } \text{Re}\left[\frac{1}{\text{sigmaR}^2}+\frac{1}{\text{beta}-\text{Deltau}-\text{tau}}\right]>\text{Re}\left[\frac{1}{\text{beta}}\right]$$

Here we don't care about the constants, but now add to the exponent the first term in eq. (14) that does not depend on x_β. Then we organize in powers of $x_{\tau+\Delta\tau}$

In[•]:=

```
Series[(sigmaR^2 (xtauplus - xzero)^2 - (Deltau + tau) xzero^2 + beta (-xtauplus^2 + xzero^2)) /
   (2 (beta^2 - beta (Deltau + tau) + sigmaR^2 (Deltau + tau))) -
  (xtau - xtauplus)^2 / (2 Deltau), {xtauplus, 0, 4}]
```

Out[•]=

$$\left(-\frac{\text{xtau}^2}{2\,\text{Deltau}}+\frac{\text{beta}\,\text{xzero}^2+\text{sigmaR}^2\,\text{xzero}^2-(\text{Deltau}+\text{tau})\,\text{xzero}^2}{2\left(\text{beta}^2-\text{beta}\,(\text{Deltau}+\text{tau})+\text{sigmaR}^2\,(\text{Deltau}+\text{tau})\right)}\right)+$$
$$\left(\frac{\text{xtau}}{\text{Deltau}}-\frac{\text{sigmaR}^2\,\text{xzero}}{\text{beta}^2-\text{beta}\,(\text{Deltau}+\text{tau})+\text{sigmaR}^2\,(\text{Deltau}+\text{tau})}\right)\text{xtauplus}+$$
$$\left(-\frac{1}{2\,\text{Deltau}}+\frac{-\text{beta}+\text{sigmaR}^2}{2\left(\text{beta}^2-\text{beta}\,(\text{Deltau}+\text{tau})+\text{sigmaR}^2\,(\text{Deltau}+\text{tau})\right)}\right)\text{xtauplus}^2+O[\text{xtauplus}]^5$$

The term in front of the $(x_{\tau+\Delta\tau})^2$, up to a factor of two, is the inverse variance of the one-step Gaussian, but that we can simplify a bit.

In[•]:=

$$\textbf{FullSimplify}\left[1\Big/\left(-\frac{1}{\text{Deltau}}+\frac{-\text{beta}+\text{sigmaR}^2}{\left(\text{beta}^2-\text{beta}\,(\text{Deltau}+\text{tau})+\text{sigmaR}^2\,(\text{Deltau}+\text{tau})\right)}\right)\right]$$

Out[•]=

$$\frac{\text{Deltau}\left(-\text{beta}^2+\text{beta}\,(\text{Deltau}+\text{tau})-\text{sigmaR}^2\,(\text{Deltau}+\text{tau})\right)}{\text{beta}^2-\text{beta}\,\text{tau}+\text{sigmaR}^2\,\text{tau}}$$

This is eq. (16), for the standard deviation of the Gaussian.

The mean value of the Gaussian is given by the ratio of the linear to the square term.

In[∘]:=

```
FullSimplify[(xtau/Deltau - (sigmaR^2 xzero)/(beta^2 - beta (Deltau + tau) + sigmaR^2 (Deltau + tau))) /
  (1/Deltau + (beta - sigmaR^2)/(beta^2 - beta (Deltau + tau) + sigmaR^2 (Deltau + tau)))]
```

$$\text{Out}[\circ]= \frac{\text{beta}^2\ \text{xtau} - \text{beta}\ (\text{Deltau} + \text{tau})\ \text{xtau} + \text{sigmaR}^2\ ((\text{Deltau} + \text{tau})\ \text{xtau} - \text{Deltau}\ \text{xzero})}{\text{beta}^2 - \text{beta}\ \text{tau} + \text{sigmaR}^2\ \text{tau}}$$

This is eq. (15) for the mean value of $x_{\tau+\Delta\tau}$.